

1) Anonymous Pipes:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cd ..  
himasree@LAPTOP-41COT5AA:~/OSLab$ ./Apipes  
Child received: Hello from Parent!  
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```
#include <stdio.h>  
#include <unistd.h>  
#include <string.h>  
#include <sys/types.h>  
int main()  
{  
    int fd[2];  
    pid_t pid;  
    char message[] = "Hello from Parent!";  
    char buffer[100];  
    // Create pipe  
    pipe(fd);  
    // Create child process  
    pid = fork();  
    if (pid > 0)  
    {  
        // Parent process  
        close(fd[0]); // Close reading end  
        write(fd[1], message, strlen(message) + 1);  
        close(fd[1]); // Close writing end  
    }  
    else if (pid == 0)  
    {  
        // Child process  
        close(fd[1]); // Close writing end
```

```
        // Child process  
        close(fd[1]); // Close writing end  
  
        read(fd[0], buffer, sizeof(buffer));  
        printf("Child received: %s\n", buffer);  
        close(fd[0]); // Close reading end  
    }  
    return 0;  
}
```

2) Named Pipes / FIFO:

a) Create a FIFO

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano FIFO_CREATE.c  
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc FIFO_CREATE.c -o FIFO_CREATE  
himasree@LAPTOP-41COT5AA:~/OSLab$ ./FIFO_CREATE  
mkfifo: File exists  
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```

#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>

int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");

    return 0;
}

```

b) FIFO_READ

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("myfifo", O_RDONLY);

    int n = read(fd, buffer, sizeof(buffer) - 1);
    buffer[n] = '\0';

    printf("Received: %s\n", buffer);

    close(fd);

    return 0;
}

```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ ./FIFO_READ
Received: Hello from Writer

```

c. FIFO_write.c

```

himasree@LAPTOP-41COT5AA:~$ cd ~/OSLab
himasree@LAPTOP-41COT5AA:~/OSLab$ ./FIFO_WRITE
Message sent
himasree@LAPTOP-41COT5AA:~/OSLab$

```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd;
    char message[] = "Hello from Writer";

    fd = open("myfifo", O_WRONLY);

    write(fd, message, strlen(message) + 1);

    close(fd);

    printf("Message sent\n");

    return 0;
}
```